Talent Carriage

Azure capacity and fixed cost

AI Attendance Agent: what the current setup can handle, and what to upgrade, at what fixed monthly cost, as customers grow.

$74 / month	2–3 calls	~500 / hour
Fixed Azure cost today (~INR 6,500)	Phone calls the platform can hold at once today	WhatsApp conversations it can handle today

Scope. Azure only, at Microsoft's public list price for Central India, before GST, in US dollars (INR at ~88 per dollar). Not included: Plivo, LiveKit, WhatsApp (Meta), Sarvam, and Azure OpenAI tokens, which are billed per use and have no fixed charge. Figures are estimates: this login cannot read the Azure bill.

At a glance

Step	Built for	Calls at once	WhatsApp chats / hour	Fixed Azure cost / month
Today	Pilots: a few small companies	2–3	~500	~$74 (~INR 6,500)
Step 1	One 10,000-employee company, or ~10 companies of 1,000	~15	~1,200	~$310 (~INR 27,000)
Step 2	One 100,000-employee company	~150	~3,600	~$950 (~INR 84,000)
Step 3	One 1,000,000-employee company, or ~10 of 100,000	~1,250	~28,000	~$4,850 (~INR 4.3 lakh)

The three biggest levers, in order: **(1)** two code fixes that every step needs (send contacts in parallel through the job queue, and stop holding a database connection while the AI thinks); **(2)** the voice worker's size and copy count; **(3)** the Azure OpenAI quota. Spreading contacts across the working day instead of one morning burst cuts every peak figure in this document by about four.

Companies each step can serve

Assuming every unexplained day gets a phone call. If most are settled on WhatsApp, each step serves roughly twice as many.

Company size	Cases a day	Today	Step 1	Step 2	Step 3
100 employees	5	~20	~120	~1,200	~10,000
1,000 employees	50	~2	~12	~120	~1,000
10,000 employees	500	0	1	~12	~100
100,000 employees	5,000	0	0	1	~10
1,000,000 employees	50,000	0	0	0	1

1. The current setup

Subscription "Azure subscription 1", tenant Default Directory, resource group *rg-client-hub*, read with the Azure CLI on 27 Oct 2026.

Resource	Azure name	Size	$ / month
Web app (portal, webhooks, call turns, job runner)	tc-attendance-plan / tc-attendance-99772	App Service B1 Linux, 1 instance, Always On	13.1
Database	tc-clienthub-pg-99772	PostgreSQL Flexible B1ms, 32 GB, 7-day backups, no HA. Shared with Client Hub	22.1
Voice worker (LiveKit agent)	tc-attendance-env / tc-attendance-worker	Container Apps, 0.5 vCPU / 1 GiB, always 1 copy	~34
Container registry	tcattendanceacr99772	Basic, ~2.2 GB of 10 GB included	5.1
Call recordings	tcattendancerec99772	Storage, Hot LRS, ~27 MB	<0.1
Azure OpenAI	tc-attendance-aoai (South India)	gpt-4.1-mini, text-embedding-3-small, gpt-4o-transcribe, pay per use	usage only
Total			**~74**

The voice worker's figure assumes it is billed at the active rate (its CPU averaged ~47 millicores over 7 days, above Azure's idle threshold). Billed as idle it would be ~$10, bringing the total to ~$50. Splitting the shared database evenly with Client Hub gives this platform a share of ~$63.

What it can do today

Limit	What sets it	Capacity
Calls at the same time	Voice worker: 0.5 vCPU / 1 GiB, maximum 1 copy. Each call runs its own speech-detection process.	2–3 calls
AI throughput	gpt-4.1-mini quota: 200K tokens and 200 requests a minute, shared by calls and WhatsApp	~20 calls at once, or ~500 chats an hour
Replies being worked on at once	Each AI reply holds 1 of the web app's 20 database connections while the AI thinks	~20 per web server
Database connections	Server allows 50, shared with Client Hub	Blocks adding more than ~2 web servers
Sending contacts	Sent one by one inside the HR person's click; App Service stops a request at 230 seconds	~1 a second, ~200 per click

How the estimates were made

- About 5% of employees have an unexplained day on a working day (absence, half day, missed punch).
- Contacts go out after the morning upload, so most chats and calls land in a ~2-hour peak, on ~22 working days a month.
- A call lasts ~3 minutes with ~10 AI turns (~30K tokens). A WhatsApp chat has ~4 replies (~25K tokens).
- One voice-worker vCPU holds about 5 calls. Calls at peak = cases ÷ 40 when everyone is called.
- Prices come from the Azure Retail Prices API for Central India (South India for Azure OpenAI), at 730 hours a month. Reserved instances (1 or 3 years) cut App Service, database and Container Apps compute by roughly 30–60%.
- Container Apps figures subtract the monthly free grant and assume active-rate billing. Scaled-out copies are charged only for peak hours.
- Token usage per call and per chat is estimated from the code, not measured. The cost boxes under each transcript record real usage and should be used to refine these figures once there is volume.
- This login has no access to Azure Cost Management, so the current-setup total is a list-price estimate, not the billed amount.

2. What each company size needs at peak

Employees	Cases a day	Calls at once (all called)	WhatsApp chats in the peak	AI tokens / minute	Voice worker vCPU
100	5	under 1	5	~1K	under 1
1,000	50	1–2	50	~15K	under 1
10,000	500	~13	500	~130K	~3
100,000	5,000	~125	5,000	~1.25M	~25
1,000,000	50,000	~1,250	50,000	~12M	~250

Do these first, before any step. These two code changes cost nothing on Azure and lift the biggest software limits:
• Send contacts through the background job queue, several at a time, instead of one by one inside the click (*src/app/actions.ts*, *src/lib/detection/run.ts*).
• Release the database connection before calling the AI, in WhatsApp replies (*api/webhook*) and call turns (*api/agent/turn*).

3. Step 1: up to one 10,000-employee company

For one company of 10,000 employees, or about 10 companies of 1,000, all called on the same morning.

Capacity: ~15 calls at once and ~1,200 WhatsApp chats an hour.

Resource	Today	Upgrade to	$ / month
Voice worker	0.5 vCPU / 1 GiB, 1 copy	2 vCPU / 4 GiB per copy, 1 to 3 copies, CPU scale rule. Extra copies run only at peak	~171
Web app	B1 (1 core, 1.75 GB)	P0v3 (1 vCPU, 4 GB), allows autoscale later	~60
Database	B1ms, shared	B2s (2 vCPU, 4 GB), 32 GB storage, still shared with Client Hub	~76
Azure OpenAI	200K tokens / min	Raise gpt-4.1-mini quota to ~500K tokens / min (portal request)	0 fixed
Registry, storage	Basic, Hot LRS	No change	~6
Total	**~$74**		**~$310**

Also needed

- The two code changes above.
- Make sure the voice worker turns away new calls when it is full, so callers are not left in silence.

Approx. INR 27,000 a month.

4. Step 2: one 100,000-employee company

For one company of 100,000 employees, or about 10 companies of 10,000.

Capacity: ~150 calls at once and ~3,600 WhatsApp chats an hour.

Resource	Today	Upgrade to	$ / month
Voice worker	From Step 1	2 vCPU / 4 GiB copies, 1 always on, scaling to ~15 at peak (~44 peak hours a month)	~266
Web app	P0v3, 1 instance	P1v3 (2 vCPU, 8 GB), 2 instances, autoscale on CPU	~240
Job runner	Inside the web app	Its own Container App, 0.5 vCPU / 1 GiB, always on	~34
Database	B2s, shared	Its own server: General Purpose D4ds_v5 (4 vCore, 16 GB), 128 GB, built-in connection pooler (PgBouncer)	~383
Azure OpenAI	~500K tokens / min	~1.5M tokens / min: quota increase, or a second deployment in another region	0 fixed
Call recordings	~27 MB	~165 GB added a month; lifecycle rule moves old files to Cool	~20
Registry	Basic	No change	~5
Total			**~$950**

Also needed

- Move scheduled work into the separate job runner so the web app only serves pages and turns.
- A per-company limit on contacts per minute, so one large upload cannot starve other customers.
- Monitoring and alerts (Application Insights / Log Analytics) are recommended here; they bill per GB and are not included.

Approx. INR 84,000 a month. Client Hub keeps the old B1ms server; its cost is no longer counted here.

5. Step 3: one 1,000,000-employee company

For one company of a million employees, or about 10 companies of 100,000.

Capacity: ~1,250 calls at once and ~28,000 WhatsApp chats an hour.

Resource	Today	Upgrade to	$ / month
Voice worker	From Step 2	2 vCPU / 4 GiB copies, 2 always on, scaling to ~125 at peak. Needs a Container Apps core quota increase or dedicated workload profiles	~1,475
Web app	P1v3 × 2	P2v3 (4 vCPU, 16 GB), 3 always on, autoscale to 8	~800
Job runner	0.5 vCPU	2 vCPU / 4 GiB Container App	~152
Database	D4ds_v5	D8ds_v5 (8 vCore, 32 GB), 512 GB, standby failover (HA), plus one D4ds_v5 read replica	~2,035
AI gateway	None	API Management Basic v2 to spread AI calls across regional deployments	~150
Queue	Postgres table	Service Bus Standard	~20
Azure OpenAI	~1.5M tokens / min	~12M tokens / min across several regions	0 fixed
Recordings, registry		~1.65 TB added a month with Cool/Archive lifecycle; registry Standard	~220
Total			**~$4,850**

Also needed

- Split the largest tables (attendance_days, messages, cost_lines) into monthly partitions.
- Stagger each company's contacts across the day; without it, the peak needs about four times the voice and AI capacity.
- If pay-per-use quota cannot reach ~12M tokens a minute, reserved AI capacity (Provisioned Throughput) is needed. At list price it is about $1 per unit per hour (~$730 a month), with a minimum of roughly 15 units (~$11,000 a month). That would be the largest single cost.

Approx. INR 4.3 lakh a month, before any reserved AI capacity.

6. Limits outside Azure

Not costed here, but they cap capacity from about 10,000 employees upward and must grow with each step.

Service	What limits you	When it matters
WhatsApp (Meta)	Daily cap on conversations you start: 1K, then 10K, 100K, unlimited per 24 hours, raised by sending volume and quality	From ~20,000 employees
Plivo	Calls per second you can place, and concurrent channels on the account	From Step 1
LiveKit	Concurrent sessions allowed by the plan	From Step 1
Sarvam	Speech-to-text and text-to-speech rate limits	From Step 2